

CDMA

Each User is assigned an orthogonal code.

Users: — 11101000 111-11-1-1-1

User 1 → 1 Data x Code

1 x 11101000

111-11-1-1-1 → time slot 0

User → 0 → (1 x 111-11-1-1-1)

-1-1-11-111 → time slot 1

Receiver

x 111-11-1-1-1
111-11-1-1-1

Σ 111 1 11 1 1

= +8 > 0 → 1

-1-1-1 1-111
1 1 1 -1 1-1-1

Σ -1-1-1-1-1-1-1-1
= -8 → 0

User1 $\Rightarrow$ 1 1 1 -1 1 -1 -1 -1

User2 $\Rightarrow$ 1 -1 1 1 1 -1 1 1

User1 $\rightarrow$ time slot 0 $\rightarrow$ 1
1 $\rightarrow$ 0

2 $\rightarrow$ time slot 0 $\rightarrow$ 1
1 $\rightarrow$ 1

User1 time slot 1
+
User2
[2 0 2 0 2 -2 0 0]

Receiver
Side
At time slot
0

Receiver to Receiver user 1's data will
combine with user 1's code

2 0 2 0 2 -2 0 0
1 1 1 -1 1 -1 -1 -1
2 0 2 0 2 2 0 0

tg

Receiver to receive user 2's data will
combine with user 2's code

$$\begin{array}{cccccccc}
 2 & 0 & 2 & 0 & 2 & -2 & 0 & 0 \\
 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 \\
 \hline
 2 & 0 & 2 & 0 & 2 & 2 & 0 & 0
 \end{array}$$

tg

line slot 2

$$\begin{array}{l}
 \text{User 1} \rightarrow 0 \quad (1 \ 1 \ 1 \ -1 \ 1 \ -1 \ -1 \ -1) \\
 \text{User 2} \rightarrow 1 \quad (1 \ -1 \ 1 \ 1 \ 1 \ -1 \ 1 \ 1)
 \end{array}$$

$$\begin{array}{l}
 \text{User 1} \rightarrow -1 \ -1 \ -1 \ 1 \ -1 \ 1 \ 1 \ 1 \\
 \text{User 2} \rightarrow 1 \ -1 \ 1 \ 1 \ 1 \ -1 \ 1 \ 1
 \end{array}$$

Combined Signal

$$\begin{array}{cccccccc}
 0 & -2 & 0 & 2 & 0 & 0 & 2 & 2 \\
 \hline
 -3 & -3 & -3 & 3 & -3 & 3 & 3 & 3 \\
 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1
 \end{array}$$

User 1's power = 3x User 2's power.

Receiver to receive user 1's data
multiplies combined signal with user 1's
code

$$\begin{array}{r} 0 \quad -2 \quad 0 \quad 2 \quad 0 \quad 0 \quad 2 \quad 2 \\ 1 \quad 1 \quad 1 \quad -1 \quad 1 \quad -1 \quad -1 \quad -1 \end{array}$$

$$\sum 0 \quad -2 \quad 0 \quad -2 \quad 0 \quad 0 \quad -2 \quad -2$$

$$= -8$$

User 2

$$\begin{array}{r} 0 \quad -2 \quad 0 \quad 2 \quad 0 \quad 0 \quad 2 \quad 2 \\ 1 \quad -1 \quad 1 \quad 1 \quad 1 \quad -1 \quad 1 \quad 1 \end{array}$$

$$\sum 0 \quad 2 \quad 0 \quad 2 \quad 0 \quad 0 \quad 2 \quad 2$$

$$+ 8 \quad \checkmark \quad \parallel$$